

More risk Pooling

Problem

- SY Sporting Apparel prints up novelty T-shirts commemorating major sport events (e.g. Super Bowl, the World Series, Lakers winning the NBA Tournament). The T-shirts (irrespective of the event) cost \$5 to produce and distribute, and they sell for \$20. Company policy is to donate, after any event, all excess inventories, to the Salvation Army. Ignore goodwill loss due to unmet demand. SY is planning to produce shirts for the NBA finals. The demand for the shirts is uncertain due to the uncertainty in the performance of the home team. The mean demand is 100000 and the standard deviation is 30000. You may assume that the demand is normally distributed,

1. How many shirts should SY manufacture?
2. What is the resulting service level
3. How would you change your answer to **(1)** if instead of donating the shirts to the Salvation Army the company decides to sell the excess inventory at \$2 per shirt? (You may assume that the company can sell all excess inventory at that price)

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$$r = 20 \quad c = 5$$

$$\text{Service level} = (r - c) / r = 15 / 20 = 75\% \text{ or } 0.75$$

How much to order?

- To cover 75% of demand, we need to find out how many standard deviations from the mean we need to cover. (This logic works because we have a Normally distributed demand).
- You can do `NORMSINV(0.75)` in excel which will give you 0.68

OR if you want to use Old fashioned Normal tables (pls see next slide)

The number of standard deviations from the average that corresponds to 0.75 service level is therefore 0.68!

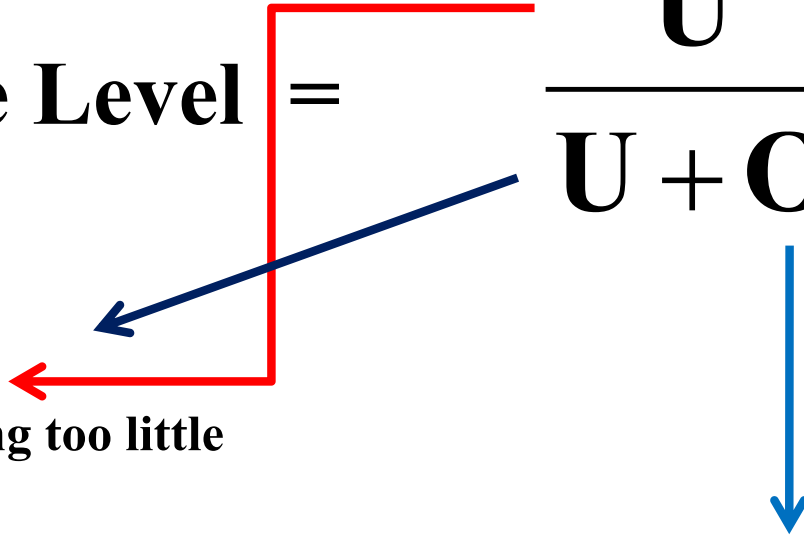
therefore, you would add 0.68 times the standard deviation to the average, which should give you $100,000 + 0.68 \times 30,000 = 118,400$ units.

3. How would you change your answer to (1) if instead of donating the shirts to the Salvation Army the company decides to sell the excess inventory at \$2 per shirt? (You may assume that the company can sell all excess inventory at that price).

Change in service level

- $r=20$ $c=5$
- How does the service level change when you can now sell the excess at \$2.
- To see this, pls interpret the service level using overage and underage costs that we discussed in lecture 6.
 - By selling the excess at \$2, the firm is essentially decreasing the cost of overstocking (O) by \$2.

Recall: Intuition behind optimal Service Level expression

$$\text{Optimal Service Level} = \frac{U}{U + O}$$


The per unit Underage cost
That is the Per unit cost of having too little
inventory

The per unit Overage cost
That is the Per unit cost of having too much
inventory

Change in service level

- Writing as overage O and underage U
- $U = r - c = \$15$
- $O = \$5 - \$2 = \$3$
- Therefore, Optimal service level equal =
 $15 / (15 + 3) = 15 / 18 = 0.8333$

How many standard deviations from the mean corresponds to 0.833. Well, u need to use Normsinv in excel or use as a Normal table (see next page)

Normsinv(0.833)is 0.96

Order quantity

$$\begin{aligned} Q &= \text{Average} + 0.96 * \text{standard deviation} \\ &= 100000 + .96 * 30000 = 128800. \end{aligned}$$

The order quantity is higher than before as the service level has gone up. The service level has gone up because the cost of overstocking decreased due to the fact excess can now be sold at \$2.

Another Problem

SY decides to pursue an alternate plan. The NBA Finals is in June and the Super bowl is in January of every year. SY decides to come up with a forecast (of the T-shirt demand) in December for these two events and to manufacture white shirts by the end of December. Then, as the events unfold, SY will dye and print the white T-shirts with the details of the events at a later date much closer to the demand. Assume that the demand forecast in December for the NBA finals is above. If the December forecast for Super bowl is also Normally distributed with a mean = 300000 and a standard deviation of 75000, how many white T-shirts will SY produce in December if it wants to keep the same service level of 0.75 for both events.

Risk Pooling

- Service level is 0.75.
- Two forecasts --- independent demands
 - N (300000, 75000) Super bowl
 - N(100000, 30000) NBA
- Combined forecast
 - Mean = 300000+100000 = 400000
 - Stddev = $\sqrt{(30000)^2 + (75000)^2} = 80777$
 - New combined forecast: N(400000, 80777)

Order quantity

$$\begin{aligned} Q &= \text{Mean} + .68 * \text{Sted dev} = \\ &400000 + 0.68 * 80777 \\ &= \mathbf{454928 \text{ Units}} \end{aligned}$$

If SY had produced white T shirts independently at 75% service levels for the two events without combining them, the number of T shirts would have been higher. I hope you can relate what is happening here to the Benetton example we did in class.

Problem

Straight-Back Inc., manufactures high-end chairs that are sold for \$400 per chair and cost \$300 to make. Past data shows that the average monthly demand is 1000 with a standard deviation of 200 chairs. The normal distribution is a reasonable fit. The company imputes an excess inventory cost of \$5 per unsold chair at the end of the month. The management is cognizant of the fact that the market for high-end chairs is very finicky and a lost sale is costly in the long run. Based on this fact, the company decided to provide 95% serviceability.

Questions

1. What is the imputed cost of goodwill?

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$C = \$300$ $r = \$400$ Excess Inv Cost = \$5

Optimal service level = 95%

$O = \text{Overage Cost} = 300 + 5$

$U = r + \text{Goodwill (g)} = 400 + g$

Optimal service level = $U / (U + O)$

Questions

1. What is the imputed cost of goodwill?

$C = \$300$ $r = \$400$ Excess Inv Cost = \$5

Optimal service level = 95%

To find the value of goodwill = g .

$$(400+g-300) / [(400+g-300) + (300+5)] = 0.95$$

Solving gives = \$5695

Questions

2. What is the optimal base stock?

$$Q = \mu + z\sigma = 1000 + 1.65 * 200 = 1000 + 330 = 1330$$

Design question

When firms design their supply chains, they trade-off the customer needs that are to be met with the cost of meeting these needs. Two crucial cost measures that factor into design are (a) the cost of inventory and (b) the transportation costs. Transportation costs, roughly speaking can be broken down into two pieces, i.e., *inbound costs* (cost of bringing materials into a facility) and *outbound costs* (cost of sending materials out of a facility). Think of any reasonable firm (amazon.com, Zara etc.) who toy between opening up more distribution warehouses versus serving customers from fewer such facilities. Please explain clearly the pros and cons in this trade-off. In particular, pay attention to inventory costs, transportation costs, facility costs and explain what happens to each one of these with sound reasoning behind your analysis.

Bull whip effect

- (1) if I were to tell you a supply chain suffers from bull whip, how would you quantify this effect?
- (2) list some reasons why Barilla's supply chain suffered from bull whip?

BEST OF LUCK!